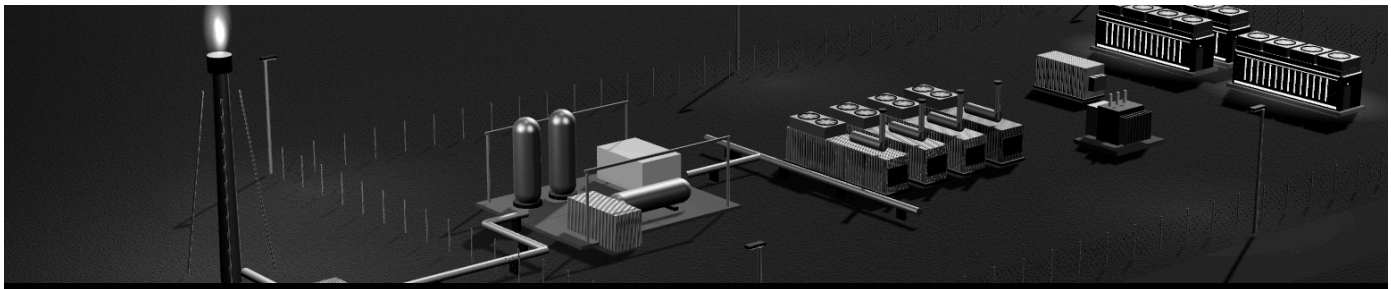
Lake County Solid Waste Facility, Painesville, Ohio: a first sizing

What the gas the site now flares could power, how much land it would take, and what it could earn each year.

3.3 MW	3	9 ac	302k
POWER FROM THE GAS	COMPUTE MODULES	LAND NEEDED	MMBTU BOUGHT / YR

GAS SALE, PER YEAR	LAND LEASE, PER YEAR	TOTAL TO THE OWNER, PER YEAR
up to $226k	$9k to $18k	$9k to $244k



YOUR SITE IN OUR LIVE 3D MODEL — ACCRETIONCOMPUTE.COM/SITE-MODEL?SRC=LFG&FLOW=1157&CH4=52

THE MATH

Landfill gas collected and flared (2022)	1,157 scfm
Methane content (2022)	51.6%
Energy in the gas	36.3 MMBtu per hour
Power at a heat rate of 11,000 Btu/kWh	about 3.3 MW
Compute (power less cooling and losses, /1.2)	about 2.7 MW
Compute modules (about 1 MW each)	3
Gas engines (running plus one spare)	4 at 1.6 MW
Land: 5 acres plus 1.2 acres per MW	about 9 acres
Gas bought per year (95% run time)	302k MMBtu
Gas payment, up to $0.75/MMBtu	up to $226k
Land lease at $1,000 to $2,000 per acre per year	$9k to $18k

Landfill gas: about 1,012 Btu per cubic foot of methane. Prices in US dollars; the price band is the one shown on our Site model page.

WHAT IT WOULD LOOK LIKE

- Gas engines on a small parcel near the gas, a battery to smooth the load, and factory-built compute modules leased to an AI company.
- No grid connection and no export: all of the power is used on site.
- We build and run the site. The owner puts in no capital.
- Multi-year gas and land agreements, matched to the tenant's lease, at a fixed price.

WORTH CHECKING

- Status of the County's 2026 landfill gas RFQ
- Latest flow and methane data, and the outlook to closure (2048)
- Space for a small fenced parcel near the flare
- Gas quality (siloxanes, H2S) for engines
- Painesville Township zoning for the parcel

Source: EPA Landfill Methane Outreach Program (LMOP) database, September 2024 release (2022 data): about 1.67 million cubic feet a day collected and flared; earlier boiler projects shut down by 2020. This is a first read from public data, not an offer: no cost and no obligation. The real price and terms are set in the gas and land agreements after a site review.